Chapter 17

Co-word and Keyword Co-occurrence

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17.1 Learning objectives

After completing this chapter, you will be able to:

- Explain the principles of co-word analysis and its role in science mapping
- Extract and clean keywords from OpenAlex data (author keywords and concepts)
- Build a keyword co-occurrence matrix and convert it to a network
- Apply community detection to identify topical clusters
- Visualise keyword co-occurrence networks with interpretable layouts

17.2 Setup

```
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(glue)
```

```
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

17.3 Conceptual background

Co-word analysis maps the conceptual structure of a research field by examining which terms appear together in the same documents. Introduced by Callon et al. (1983), the method assumes that when two keywords repeatedly co-occur across publications, they are thematically related. The resulting co-occurrence network reveals the topical landscape of a field: clusters of densely connected keywords represent coherent research themes, while bridges between clusters indicate interdisciplinary connections.

The method works with several types of terms:

- **Author keywords**: Terms selected by the paper's authors. These are intentional descriptors but suffer from inconsistency — authors may use different terms for the same concept ("machine learning" vs. "ML" vs. "statistical learning").
- **Indexed keywords**: Terms assigned by database indexers (e.g., MeSH terms in PubMed). More consistent but only available in some databases.
- **OpenAlex concepts**: Algorithmically assigned topics at multiple levels of a hierarchical taxonomy. These provide broad coverage but may miss nuanced distinctions (Priem et al., 2022).
- **Title/abstract words**: Extracted via text mining. Comprehensive but noisy; requires extensive preprocessing.

Normalisation is important for co-word networks. Raw co-occurrence counts favour high-frequency terms. Common normalisations include the **association strength** (equivalent to pointwise mutual information) and the **equivalence index** (cosine of the co-occurrence vector). Waltman et al. (2010) demonstrated that association strength produces the most balanced network structures for bibliometric mapping.

Co-word analysis complements citation-based methods (16.3). Citation networks reveal intellectual influence; co-word networks reveal thematic content. Combining both provides a more complete picture of a field's structure.

17.4 Worked example

17.4.1 Extracting keywords

We use OpenAlex-assigned topic labels as terms in a sample of scientometrics research. These are algorithmic labels, not author-supplied keywords. The flattened topic table also includes parent subfields, fields, and domains; retain only topic identifiers (`T...`) and count each term once per work.

```
works <- fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2019-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)
```

```
keywords <- works |>
  select(id, topics) |>
  unnest(topics, names_sep = "_") |>
  filter(str_detect(topics_id, "(^|/)T[0-9]+$")) |>
  filter(topics_display_name != "") |>
  select(work_id = id, keyword = topics_display_name) |>
  mutate(keyword = str_to_lower(str_trim(keyword))) |>
  filter(!is.na(keyword), nchar(keyword) >= 3) |>
  distinct(work_id, keyword)

kw_counts <- keywords |>
  count(keyword, sort = TRUE)

cat(glue("Total keyword occurrences: {nrow(keywords)}\n"))
```

```
#> Total keyword occurrences: 1500
```

```
cat(glue("Unique keywords: {n_distinct(keywords$keyword)}\n"))
```

```
#> Unique keywords: 136
```

```
kw_counts |>
  head(20) |>
  mutate(keyword = fct_reorder(keyword, n)) |>
  ggplot(aes(x = n, y = keyword)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Frequency", y = NULL) +
  theme_sci()
```

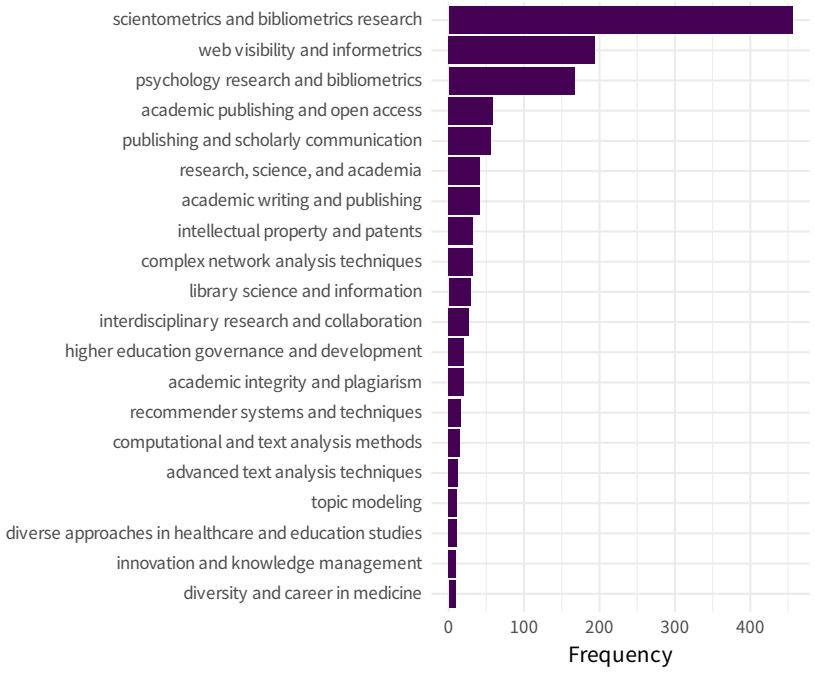


Figure 17.1: Top 20 most frequent assigned topics used as terms in the Scientometrics sample.

17.4.2 Building the co-occurrence network

We create edges between keywords that appear in the same paper.

```
kw_frequent <- kw_counts |>
  filter(n >= 5) |>
  pull(keyword)

kw_filtered <- keywords |>
  filter(keyword %in% kw_frequent)

coword_pairs <- kw_filtered |>
  inner_join(kw_filtered, by = "work_id", suffix = c("_a", "_b"),
             relationship = "many-to-many") |>
  filter(keyword_a < keyword_b) |>
  count(keyword_a, keyword_b, name = "cooccurrence")

coword_top <- coword_pairs |>
  filter(cooccurrence >= 3)

g_kw <- graph_from_data_frame(
  coword_top |> select(keyword_a, keyword_b, weight =
 ↪  cooccurrence),
  directed = FALSE
) |>
  simplify(edge.attr.comb = list(weight = "sum"))

cat(glue("Keyword network: {vcount(g_kw)} nodes, {ecount(g_kw)}
 ↪  edges\n"))
```

```
#> Keyword network: 34 nodes, 66 edges
```

17.4.3 Community detection for topical clusters

```
V(g_kw)$degree <- degree(g_kw)
V(g_kw)$strength <- strength(g_kw)

communities <- cluster_leiden(g_kw, resolution_parameter = 0.8,
                              objective_function = "modularity")
V(g_kw)$community <- as.factor(membership(communities))

cat(glue("Communities:
 ↪  {length(unique(membership(communities)))}\n"))
```

```
#> Communities: 5
```

```
cat(glue("Modularity: {round(modularity(g_kw,
 ↪  membership(communities)), 3)}\n"))
```

```
#> Modularity: 0.051
```

```
community_summary <- tibble(
  keyword = V(g_kw)$name,
  community = V(g_kw)$community,
  strength = V(g_kw)$strength
) |>
  group_by(community) |>
  slice_max(strength, n = 5) |>
  summarise(top_keywords = paste(keyword, collapse = ", "),
            n_keywords = n(), .groups = "drop")

community_summary |> gt()
```

community	top_keywords	n_keywords
1	scientometrics and bibliometrics research, web visibility and informetrics, psychology research and bibliometrics, academic publishing and open access, publishing and scholarly communication	5
2	recommender systems and techniques, information retrieval and search behavior	2
3	computational and text analysis methods, advanced text analysis techniques, topic modeling	3
4	efficiency analysis using dea, innovation policy and r&d	2
5	intellectual property and patents, intellectual property law, firm innovation and growth, innovation and knowledge management	4

17.4.4 Visualisation

```
set.seed(42)
layout <- create_layout(as_tbl_graph(g_kw), layout = "fr")

ggraph(layout) +
  geom_edge_link(aes(width = weight), alpha = 0.15, colour =
   ↪  "grey60") +
  scale_edge_width_continuous(range = c(0.3, 2), guide = "none")
   ↪  +
  geom_node_point(aes(size = degree, colour = community), alpha
   ↪  = 0.8) +
```

```
  geom_node_text(aes(label = ifelse(degree > quantile(degree,
  ↪  0.85),
                                    name, NA_character_)),
                repel = TRUE, size = 2.5, max.overlaps = 20,
                 ↪  na.rm = TRUE) +
  scale_size_continuous(range = c(1, 6), guide = "none") +
  scale_colour_manual(values = palette_sci(
    n_distinct(V(g_kw)$community)
  )) +
  labs(colour = "Cluster") +
  theme_void(base_family = "sans", base_size = 11) +
  theme(legend.position = "bottom")
```

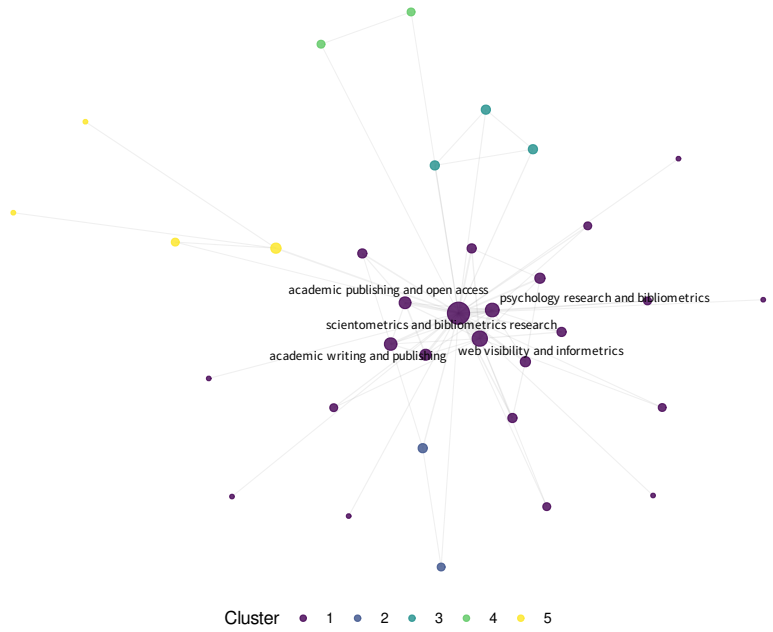


Figure 17.2: Keyword co-occurrence network coloured by topical cluster.

17.5 Diagnostics and interpretation

- **Keyword cleaning**: Inconsistent terminology inflates the vocabulary and creates spurious nodes. Standardise spelling, merge synonyms (e.g., “h-index” and “hirsch index”), and remove generic terms (“research”, “analysis”) that co-occur with everything but convey no topical information.

- **Frequency threshold**: Including rare keywords produces large, sparse, unreadable networks. Start with keywords appearing in at least 5 papers and adjust based on corpus size.
- **Community interpretability**: Each community should correspond to a recognisable research theme. If communities are uninterpretable, the resolution parameter may need adjustment or keywords need further cleaning.
- **Centrality interpretation**: High-degree keywords are thematically central (used across many contexts). High-betweenness keywords bridge distinct topics and may represent interdisciplinary concepts.

17.6 Limitations and responsible use

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- **Vocabulary inconsistency.** Author keywords are not controlled vocabulary. The same concept may appear under multiple terms, fragmenting what should be a single node. Merging synonyms requires domain expertise.
- **Algorithmic concepts.** OpenAlex concepts are assigned by machine learning and may contain errors — especially at fine-grained levels. Always validate topic assignments by reading sample papers (Priem et al., 2022).
- **Static snapshots.** A co-word network represents a time-averaged view. Emerging topics with few publications may be invisible. Consider building temporal slices to track field evolution.
- **Not evaluative.** Popular keywords indicate research activity, not research quality or societal impact. Do not equate topical centrality with importance (Hicks et al., 2015).

17.8 Common pitfalls

17.9 Common pitfalls

- **Not cleaning keywords.** Punctuation variants ("co-authorship" vs. "coauthorship"), case differences, and trailing whitespace create duplicate nodes. Clean before building the network.
- **Including stop-keywords.** Generic terms like "research", "study", "analysis" co-occur with everything and dominate the network without conveying topical information. Remove them.

- **Comparing co-occurrence counts across corpora of different sizes.** Larger corpora produce higher co-occurrence counts mechanically. Normalise or use relative measures.
- **Over-reading small clusters.** A cluster with two or three keywords may reflect a single paper, not a research theme. Report cluster size alongside content.

17.10 Exercises

1. **Author keywords vs. concepts.** Build co-occurrence networks from both author keywords (if available) and OpenAlex concepts for the same corpus. Compare the resulting topical clusters.

2. **Temporal evolution.** Split the corpus into yearly slices. For each year, build a co-word network and identify the top 5 keywords by degree. Which keywords are consistently central, and which emerge over time?

3. **Strategic diagram.** Compute density (internal cohesion) and centrality (external connections) for each keyword cluster. Plot them on a strategic diagram (density vs. centrality quadrants). Which clusters are core themes and which are peripheral?

4. **Normalisation comparison.** Build two networks from the same data: one with raw co-occurrence counts and one with association strength normalisation. How do the most central keywords differ?

17.11 Solutions

Solutions are provided in E.

17.12 Further reading

- Callon et al. (1983) — The foundational paper on co-word analysis for mapping science.
- Waltman et al. (2010) — Unified framework for bibliometric network construction, including keyword networks.
- Aria and Cuccurullo (2017) — `bibliometrix` includes co-word analysis via the `biblioNetwork()` function.
- Priem et al. (2022) — OpenAlex concepts and topic classification.

17.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
#>
#> Matrix products: default
#> BLAS:
 ↪  /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenbl
 ↪  asp-r0.3.26.so;  LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C
 ↪  LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8
 ↪  LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8      LC_NAME=C             LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8
 ↪  LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods
 ↪  base
#>
#> other attached packages:
#>  [1] ggraph_2.2.2       tidygraph_1.3.1    igraph_2.3.4
 ↪  quanteda_4.5.1
#>  [5] pdftools_3.9.1     arrow_26.0.0       bibliometrix_5.5.0
 ↪  RefManageR_1.4.0
#>  [9] bib2df_1.1.2.0     rcrossref_1.2.1    gt_1.4.0
 ↪  tidytext_0.4.3
#> [13] glue_1.8.1         openalexR_3.1.2    lubridate_1.9.5
 ↪  forcats_1.0.1
#> [17] stringr_1.6.0      dplyr_1.2.1        purrr_1.2.2
 ↪  readr_2.2.0
#> [21] tidyr_1.3.2        tibble_3.3.1       ggplot2_4.0.3
 ↪  tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.3           RColorBrewer_1.1-3
 ↪  rstudioapi_0.19.0
```

```
#>   [4] jsonlite_2.0.0         magrittr_2.0.5
 ↪  farver_2.1.2
#>   [7] rmarkdown_2.32        fs_2.1.0              vctrs_0.7.3
#>  [10] memoise_2.0.1          askpass_1.2.1
 ↪  base64enc_0.1-6
#>  [13] htmltools_0.5.9       contentanalysis_1.1.1  curl_8.1.0
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0
 ↪  htmlwidgets_1.6.4
#>  [19] tokenizers_0.3.0      plyr_1.8.9            httr2_1.3.0
#>  [22] cachem_1.1.0           plotly_4.12.1
 ↪  dimensionsR_0.0.3
#>  [25] mime_0.13              lifecycle_1.0.5
 ↪  pkgconfig_2.0.3
#>  [28] Matrix_1.7-5           R6_2.6.1
 ↪  fastmap_1.2.0
#>  [31] shiny_1.14.0           digest_0.6.39
 ↪  shinycssloaders_1.1.0
#>  [34] rprojroot_2.1.1        SnowballC_0.7.1
 ↪  labeling_0.4.3
#>  [37] urltools_1.7.3.1       timechange_0.4.0       httr_1.4.9
#>  [40] polyclip_1.10-7        compiler_4.6.1         here_1.0.2
#>  [43] bit64_4.8.6            withr_3.0.3             S7_0.2.2
#>  [46] viridis_0.6.5         ggforce_0.5.0          MASS_7.3-65
#>  [49] bibliometrixData_0.3.0 tools_4.6.1            otel_0.2.0
#>  [52] stopwords_2.3          zip_3.0.2
 ↪  rentrez_1.2.4
#>  [55] httpuv_1.6.17          promises_1.5.0         grid_4.6.1
#>  [58] stringdist_0.9.17      generics_0.1.4
 ↪  gtable_0.3.6
#>  [61] tzdb_0.5.0             rscopus_0.9.0           ca_0.72
#>  [64] data.table_1.18.6.1    hms_1.1.4              xml2_1.6.0
#>  [67] utf8_1.2.6             ggrepel_0.9.8
 ↪  pillar_1.11.1
#>  [70] later_1.4.8            tweenr_2.0.3
 ↪  lattice_0.22-9
#>  [73] bit_4.6.0              tidyselect_1.2.1
 ↪  miniUI_0.1.2
#>  [76] knitr_1.52             gridExtra_2.3.1
 ↪  bookdown_0.48
#>  [79] crul_1.6.0             xfun_0.62
 ↪  graphlayouts_1.2.5
#>  [82] DT_0.34.0              humaniformat_0.6.0
 ↪  visNetwork_2.1.4
#>  [85] stringi_1.8.9          qpdf_1.4.1            yaml_2.3.12
#>  [88] evaluate_1.0.5         httpcode_0.3.0         cli_3.6.6
#>  [91] xtable_1.8-8           systemfonts_1.3.2     Rcpp_1.1.2
#>  [94] readxl_1.6.0           triebeard_0.4.1
 ↪  XML_3.99-0.25
#>  [97] parallel_4.6.1         assertthat_0.2.1
 ↪  pubmedR_1.0.2
```

```
#> [100] viridisLite_0.4.3      scales_1.4.0
 ↪  openxlsx_4.2.9
#> [103] rlang_1.3.0            fastmatch_1.1-8
```